

CanBot.

An intelligent waste sorter that detects cans and bottles on its path using computer vision and sorts the waste using a robotic arm.



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Group #01

EE297 Project

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Hardware

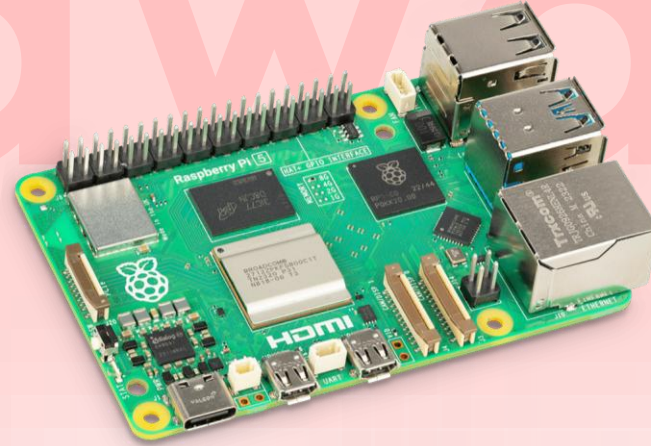
Arduino Uno

Raspberry Pi.

Logitech webcam.

Servo motors

NSC2, etc.



Softwares

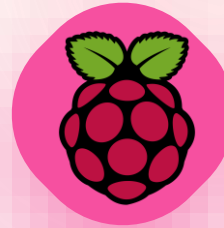


YOLO.

YOLO is a state-of-the-art, real-time object detection system

TinkerCAD.

Tinkercad is a free-of-charge, online 3D modeling program that runs in a web browser.



Raspberry Pi OS.

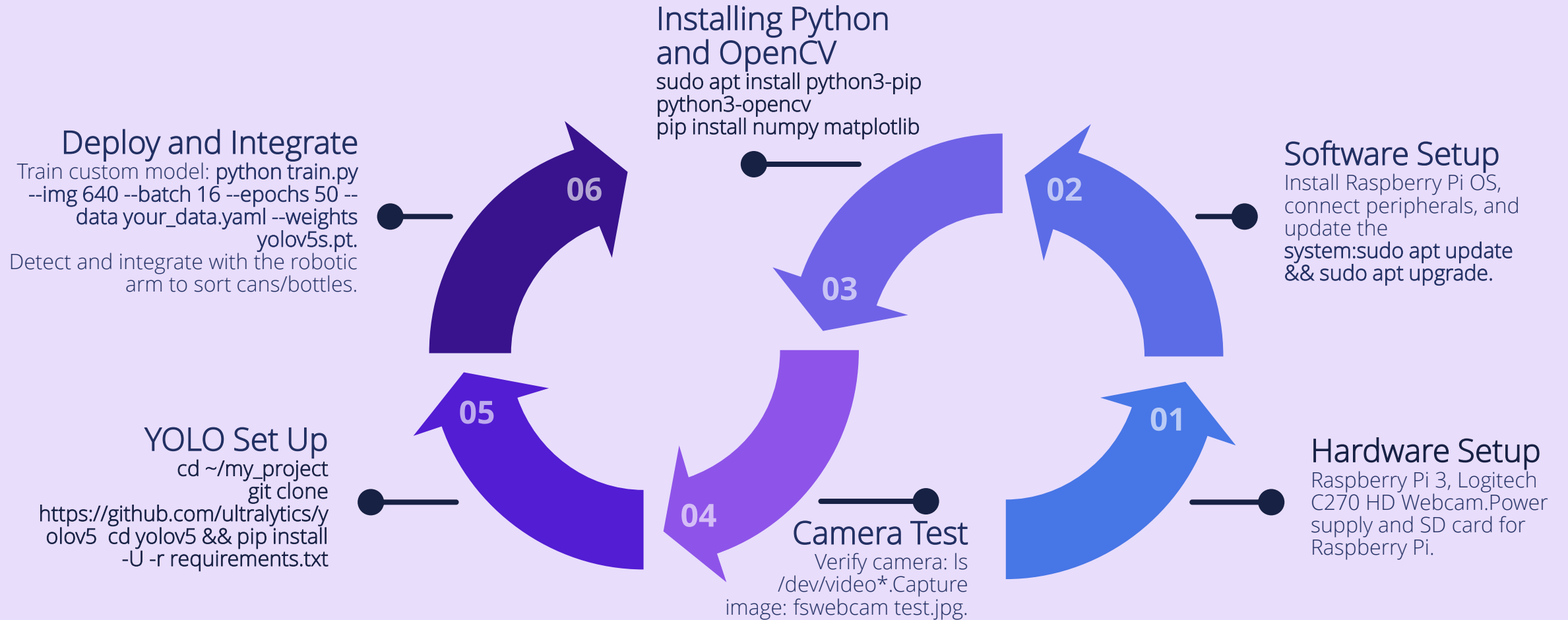
Raspberry Pi OS is a Unix-like operating system based on the Debian Linux distribution for the Raspberry Pi family of compact single-board computers.

OpenCV.

OpenCV is a library of programming functions mainly for real-time computer vision. Originally developed by Intel.



Object Detection



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Overview

Raspberry Pi: Sends data as a string.
Arduino: Decodes string into int array and controls servos

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Serial Communication

Utilizes USB port for data transmission. Ensures reliable and fast communication.

Raspberry Pi and Arduino

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Data Conversion

Raspberry Pi: Int array → String.
Arduino: String → Int array → Sets servo angles.

d

System Flow

Raspberry Pi → USB → Arduino → Servos

Purpose

Designed for precision, flexibility, and strength to handle cans, bottles, and its own weight.

Structure

Features 6 joints with 4 servos (base, 2 elbows, wrist) and 2 micro servos (wrist rotation, gripper).

Range of Motion

Full motion allows picking objects in front and placing them behind.

Future-Proofing

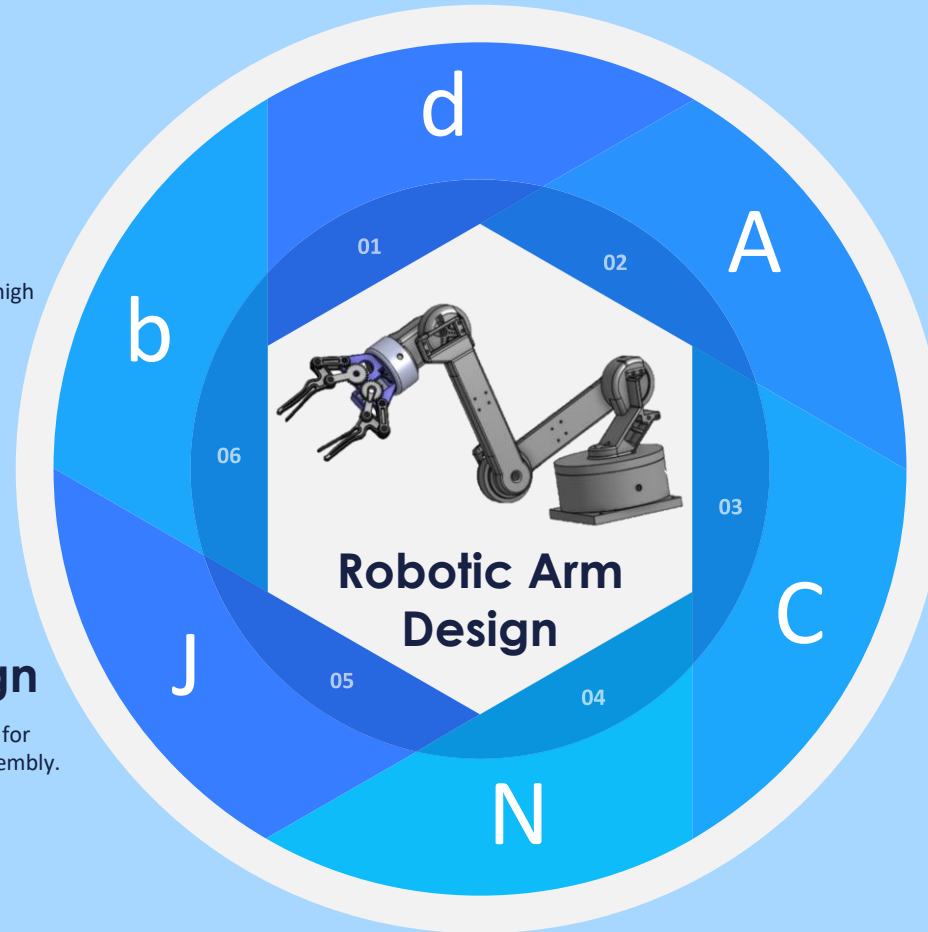
Bearing design ensures durability and adaptability.

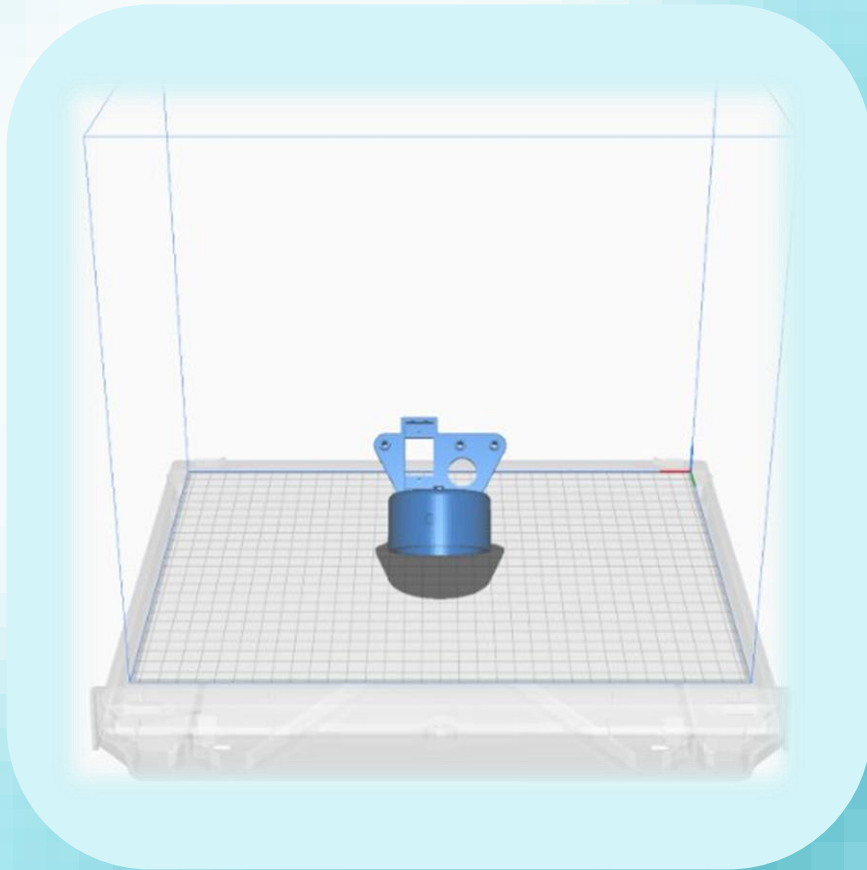
Simulation Freedom

Each link rotates $\pm 90^\circ$, offering high flexibility.

CAD Design

Modeled in SolidWorks for precise simulation and assembly.





O

3D printing.

The SolidWorks model was 3D printed in multiple parts.

Material.

Standard PLA was used for printing.

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Customisation.

3D printing provided complete design flexibility

Kinematics

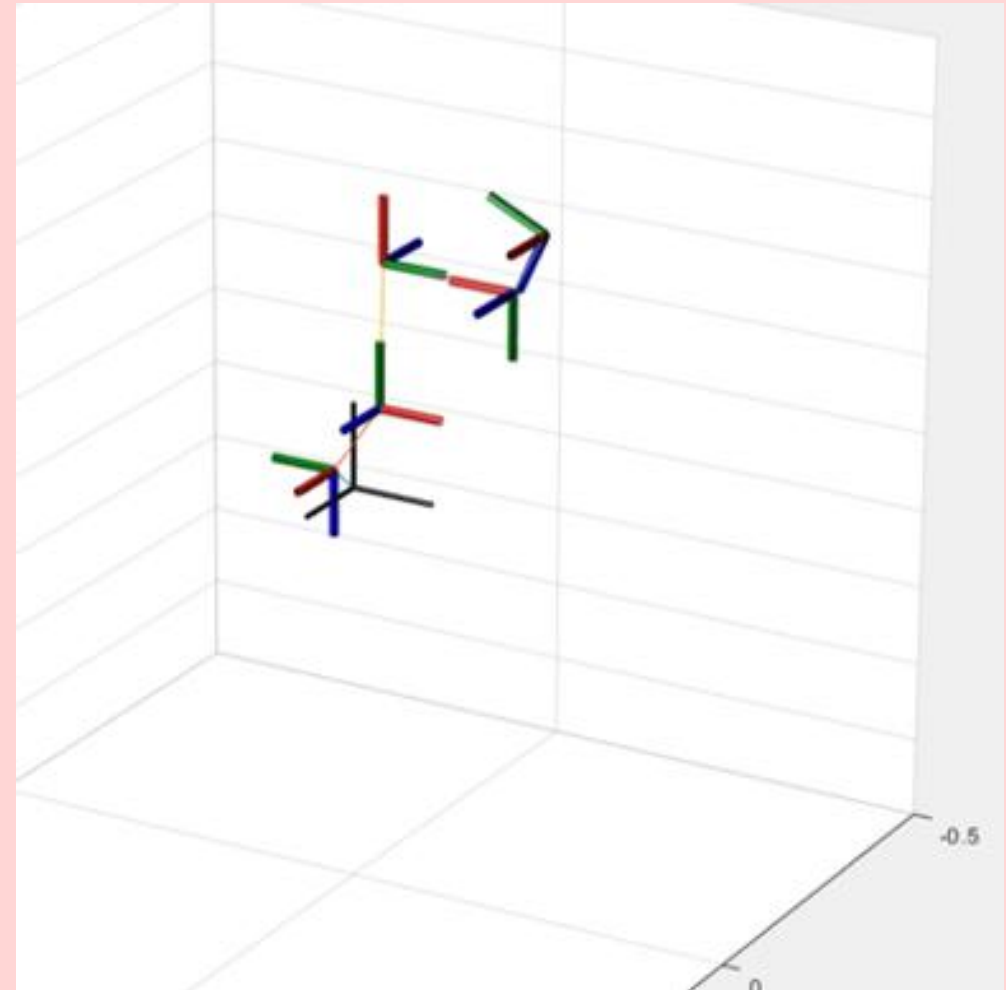
Definition: Study of motion without considering its causes.

Forward Kinematics

- Calculates the position of the robot's end-effector.
- Based on the angles of the joints in the robot arm.

Inverse Kinematics

- Determines the motion and joint angles required.
- Achieves a desired position for the end-effector.





Inverse Kinematics in use

- Controlled the robot arm using Inverse Kinematics (IK).
- Enables setting an end-effector position by inputting a coordinate.
- Computer Vision provides the coordinate of the target (e.g., can or bottle).
- The IK solver:
 - Calculates the joint angles to achieve the desired position.
 - Determines the motion required from the current position to the target.

Thanks
for watching our ppt